# Vibrations Workshop 3 - The Way of Lagrange

By: Jake Peyser

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**Resubmission Comment** (5/17/26): I did include a list of assumptions for both problems... Not sure why points were deducted. I also included new and very creative problem titles this time around.

## General Overview

As a whole, this problem set is designed to introduce energy methods (Lagrangian mechanics) with regards to deriving EOMs for various damped oscillating systems as an alternative to Newtonian mechanics.

## Problem #1 - The Disc

In problem 1, a disc with moment of inertia $J = 0.75\,\mathrm{kg\,m^2}$ is equipped with a spring attached to its edge with stiffness $k = 40\,\mathrm{kN/m}$ (at a distance $r_1 = 23\,\mathrm{cm}$ to the center) in conjunction with a dashpot with damping coefficient $R = 6\,\mathrm{kg/s}$ attached to its interior (at a distance $r_2 = 8\,\mathrm{cm}$ from the center). Given the coordinate depicted in figure 1, the system is initialized by $\theta_0 = 0.1\,\mathrm{rad}$ and $\dot{\theta}_0 = 0.01\,\mathrm{rad/s}$.

The overall objective to to derive the EOM, the coordinate (and temporal derivative) as a function of time, and to analyze the system against its parameters (e.g. frequency of oscillation).

### (a)

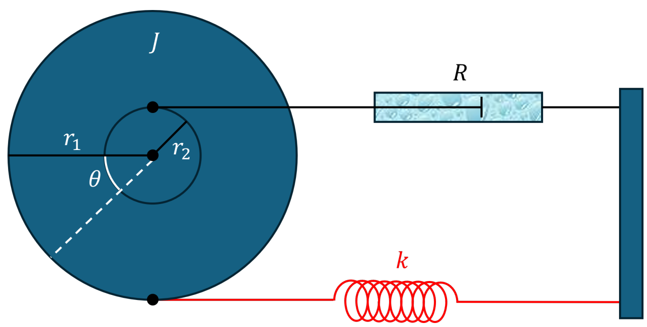


Figure 1: Damped-Oscillating Disc

Assumptions of this problem include the implicit geometry in figure 1, linear viscous damping, a quadratic (in position) spring potential, and a uniform mass distribution. Also, it will be assumed Hamilton's Principle holds.

The Lagrangian of the system is given by:

$$\mathcal{L} = \tfrac{1}{2}J\dot{\theta}^2 - \tfrac{1}{2}k(r_1 \sin\theta)^2$$

The conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = J\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -kr_1^2 \cos\theta \sin\theta$$

The damping force & lever arm:

$$\vec{F} = -R\tfrac{d}{dt}(r_2 \sin\theta, r_2) \quad \text{and} \quad \vec{r} = (r_2 \sin\theta, -r_2 \cos\theta)$$

The generalized (non-conservative) force:

$$\vec{F} \cdot \frac{\partial \vec{r}}{\partial \theta} = -Rr_2\dot{\theta}\cos\theta(1,0) \cdot (r_2\cos\theta, r_2\sin\theta) = -Rr_2^2\dot{\theta}\cos^2\theta$$

The EOM (Euler-Lagrange equation) follows as such:

$$\frac{d}{dt}\left[J\dot{\theta}\right] - \left(-kr_1^2\cos\theta\sin\theta\right) = -Rr_2^2\dot{\theta}\cos^2\theta \implies \ddot{\theta} + \frac{Rr_2^2}{J}\dot{\theta}\cos^2\theta + \frac{kr_1^2}{J}\cos\theta\sin\theta = 0$$

Upon linearizing the equation around the operating point $\theta = 0$:

$$\boxed{\ddot{\theta} + \frac{Rr_2^2}{J}\dot{\theta} + \frac{kr_1^2}{J}\theta \approx 0}$$

## (b)

The characteristic equation associated with the above ODE is given by:

$$s^2 + \frac{Rr_2^2}{J}s + \frac{kr_1^2}{J} = 0$$

This equation has roots given by (with $i^2 = -1$):

$$s = -\frac{Rr_2^2}{2J} \pm i\sqrt{\frac{kr_1^2}{J} - \left(\frac{Rr_2^2}{2J}\right)^2} = -\beta \pm i\omega_d$$

- Natural Frequency: $\omega_n = r_1\sqrt{\frac{k}{J}} = 0.23\,\mathrm{m}\sqrt{\frac{40000\,\mathrm{N/m}}{0.75\mathrm{kg\,m^2}}} \approx 53.1162\,\mathrm{rad/s}$
- Damping Coefficient: $\beta = \frac{Rr_2^2}{2J} = \frac{(6\,\mathrm{kg/s})(0.08\,\mathrm{m})^2}{2(0.75\,\mathrm{kg\,m^2})} = 0.0256\,\mathrm{s^{-1}}$
- Damped Frequency: $\omega_d = \sqrt{\frac{kr_1^2}{J} - \left(\frac{Rr_2^2}{2J}\right)^2} = \sqrt{\frac{(40000\,\mathrm{N/m})(0.23\,\mathrm{m})^2}{0.75\,\mathrm{kg\,m^2}} - (0.0256\,\mathrm{s^{-1}})^2} \approx 53.1162\,\mathrm{rad/s}$

One simple observation to make is that $\beta$ is small enough such that $\omega_n$ and $\omega_d$ agree up to 4 decimal points. That is, over small (initial) time intervals, damping is negligible.

## (c)

Provided the above quantities, the solution to the (linearized) IVP is given as such:

$$\boxed{\theta(t) = e^{-\beta t}\left(\theta_0\cos(\omega_d t) + \frac{\dot{\theta}_0 + \beta\theta_0}{\omega_d}\sin(\omega_d t)\right)}$$

$$\approx 0.1e^{-\left(0.0256\,\mathrm{s^{-1}}\right)t}\cos\left(\left(53.1162\,\mathrm{s^{-1}}\right)t - 0.0024\right)\mathrm{rad}$$

Upon (time) differentiation, one has:

$$\boxed{\dot{\theta}(t) = e^{-\beta t}\left(\dot{\theta}_0\cos(\omega_d t) - \left(\omega_d\theta_0 + \frac{\beta}{\omega_d}(\dot{\theta}_0 + \beta\theta_0)\right)\sin(\omega_d t)\right)}$$

$$\approx 5.31e^{-\left(0.0256\,\mathrm{s^{-1}}\right)t}\cos\left(\left(53.1162\,\mathrm{s^{-1}}\right)t + 1.5689\right)\mathrm{rad/s}$$

## (d)

The position ($x = \theta$ [rad]) versus time ($t$ [s]) and velocity ($v = \dot{\theta}$ [rad/s]) versus time ($t$ [s]) are plotted below. Also, the author of this writeup adapted MATLAB code from a previous project, and was admittedly too lazy to change the (axis) labeling. Apologies for the visual inconvenience.

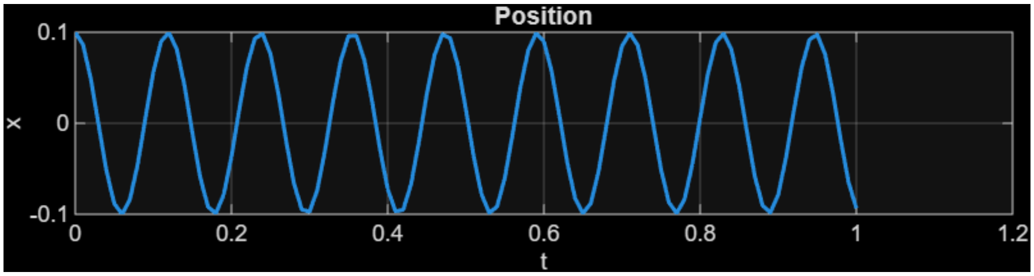


Figure 2: MATLAB Plot of $x = \theta$ [rad] Versus $t$ [s]

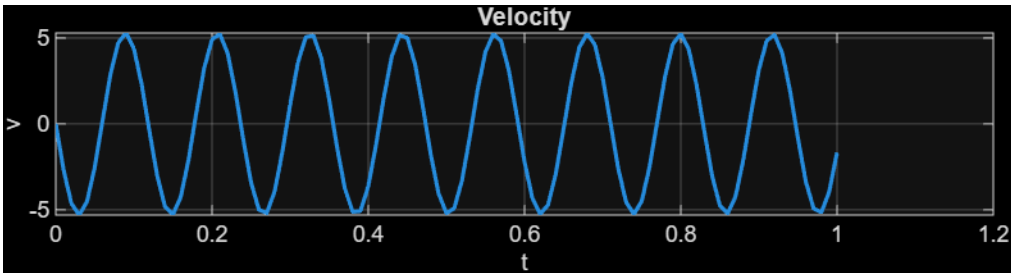


Figure 3: MATLAB Plot of $v = \dot{\theta}$ [rad/s] Versus $t$ [s]

## (e)

Although only plotted over a one second time interval, it's clear that the time it takes to decay is substantial. The envelope of the angular trajectory is given by $E_{nv}(t) = 0.1e^{-\left(0.0256\,\mathrm{s}^{-1}\right)t}\mathrm{rad}$, and so $t(E_{nv}) \approx 39.0625 \ln\left(\frac{0.1\,\mathrm{rad}}{E_{nv}}\right)\mathrm{s}$, which implies $t(1\% \text{ of } 0.1\,\mathrm{rad}) = t(0.001\,\mathrm{rad}) \approx 179.8895\,\mathrm{s}$. That is, it would take about 3 minutes for the system to settle down (99% of the way) to rest.

Otherwise, the trajectory plot seems to illustrate a sinusoid with an amplitude of 0.1 rad with a period of about 0.02 s. That's fast! The angular velocity plot is likewise sinusoidal over the depicted time interval.

## (f)

Because $\beta < \omega_d$, the system is under-damped. To make the system critically damped, it would be sufficient (fixing everything but $k$) for $\frac{kr_1^2}{J} = \left(\frac{Rr_2^2}{2J}\right)^2 \implies k = \frac{1}{J}\left(\frac{Rr_2^2}{2r_1}\right) = \frac{1}{0.75\,\mathrm{kg\,m^2}}\left(\frac{(6\,\mathrm{kg/s})(0.08\,\mathrm{m})^2)}{2(0.23\,\mathrm{m})}\right)^2 \approx 0.0093\,\mathrm{N/m}$. In other words, if the spring's stiffness $k$ were made 99.9999767713% weaker, then the system would be critically damped; any weaker would make the system over damped.

Obviously, this is a ridiculous thing to point out. However, it is demonstrative of the fact that the damping ratio is (relatively) minuscule. Heuristically, the stiffness by default is too large for the system to noticeably respond to the damping within small time intervals.

Regardless, overall, this demonstrates (mathematically) that systems with seemingly negligible damping properties inevitably bends the knee. That is, over sufficient time intervals, practically every system will come to rest when in the presence of power drains.

# Problem #2 - The Inverted Pendulum

## (a)

This problem is equivalent to the problem (d) from Workshop 2. Due to the fact an energy method was already applied there, for the sake of being heterogeneous, Newtonian mechanics will be applied here. The system is thus equipped with a FBD as depicted in figure 4.

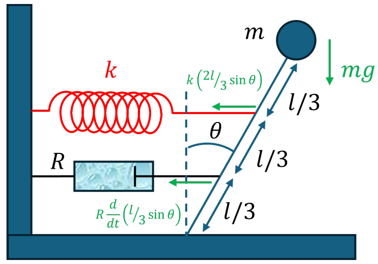


Figure 4: Inverted Pendulum with a Spring-Damping Mechanism with a FBD

Assumptions of this problem include the implicit geometry in figure 4, linear viscous damping, a quadratic (in position) spring potential, a uniform gravitational field, and a point mass distribution. Also, it will be assumed that Newton's Second Law holds.

Summing the moments about the pivot in accordance with Newton's 2nd law implies that:

$$ml^2\tfrac{d^2}{dt^2}[-\theta] = \Sigma_i M_i = [\tfrac{2}{3}l][k(\tfrac{2}{3}l\sin\theta)]\cos\theta + [\tfrac{l}{3}][(\tfrac{Rl}{3}\dot{\theta}\cos\theta)]\cos\theta - [l][mg]\sin\theta$$

$$\implies \ddot{\theta} + \frac{R}{9m}\dot{\theta}\cos^2\theta + \frac{4k}{9m}\cos\theta\sin\theta - \frac{g}{l}\sin\theta = 0$$

In accordance with the results from Workshop 2 (being that the EOM/system is identical):

$$\boxed{\omega_n = \sqrt{\frac{4k}{9m} - \frac{g}{l}}} \quad \text{and} \quad \boxed{\omega_d = \sqrt{\frac{4k}{9m} - \frac{g}{l} - \left(\frac{R}{18m}\right)^2}} = \sqrt{\frac{4k}{9m} - \frac{g}{l}}\sqrt{1 - \left(\frac{R}{18m\sqrt{\frac{4k}{9m} - \frac{g}{l}}}\right)^2}$$

Because $\omega_d = \omega_n\sqrt{1-\zeta^2}$, by direct comparison, it follows that the damping ration can be written as:

$$\boxed{\zeta = \frac{R}{18m\sqrt{\frac{4k}{9m} - \frac{g}{l}}}}$$

The only additions from the previous comments in Workshop 2 would be that the system is over damped if $\zeta > 1$, under damped if $\zeta < 1$, and critically damped if $\zeta = 1$. Also, a fair bit more thought has to go into deriving the EOM from the FBD. Lagrangian mechanics makes the derivation feel like going on autopilot from a relative standpoint.

## (b)

This problem involves a lever with a spring at one end and a dashpot at the other as depicted in figure 5. For the sake of generality, gravity, which acts at the COM of the lever, will not be neglected (for now).

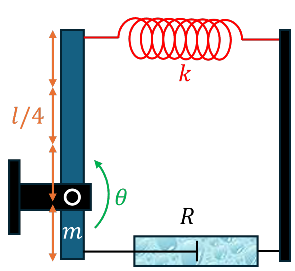


Figure 5: Lever on a Spring with Damping

Assumptions of this problem include the implicit geometry in figure 5, linear viscous damping, a quadratic (in position) spring potential, a uniform gravitational field, and a uniform mass distribution. Also, it will be assumed Hamilton's Principle holds.

The moment of inertia of the lever with respect to the pivot is calculated as such:

$$I = \int_{-l/4}^{3l/4} r^2 \frac{m}{l} dr = \left. \frac{mr^3}{3l} \right|_{r=-l/4}^{r=3l/4} = \tfrac{7}{48} ml^2$$

The Lagrangian:

$$\mathcal{L} = \tfrac{1}{2} I\dot{\theta}^2 - \tfrac{1}{2} k \left(\tfrac{3}{4} l \sin\theta\right)^2 - mg\tfrac{l}{4}\cos\theta$$

The conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = I\dot{\theta} = \tfrac{7}{48} ml^2\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -k(\tfrac{3}{4}l)^2 \cos\theta\sin\theta + \tfrac{1}{4} mgl\sin\theta$$

The damping force and (respective) position:

$$\vec{F} = -R\tfrac{d}{dt}(\tfrac{l}{4}\sin\theta, 0) \quad \text{and} \quad \vec{r} = (\tfrac{l}{4}\sin\theta, -\tfrac{l}{4}\cos\theta)$$

Generalized (non-conservative) force:

$$\vec{F} \cdot \frac{\partial \vec{r}}{\partial \theta} = -\tfrac{Rl}{4}\dot{\theta}\cos\theta(1,0)\cdot(\tfrac{l}{4}\cos\theta, \tfrac{l}{4}\sin\theta) = -R(\tfrac{l}{4})^2\dot{\theta}\cos^2\theta$$

The EOM:

$$\frac{d}{dt}\left[\tfrac{7}{48} ml^2\dot{\theta}\right] - \left(-k(\tfrac{3}{4}l)^2\cos\theta\sin\theta + \tfrac{1}{4}mgl\sin\theta\right) = -R(\tfrac{l}{4})^2\dot{\theta}\cos^2\theta$$

$$\implies \ddot{\theta} + \tfrac{3R}{7m}\dot{\theta}\cos^2\theta + \tfrac{27k}{7m}\cos\theta\sin\theta - \tfrac{12g}{7l}\sin\theta = 0$$

Upon linearlizing the EOM about the operating point $\theta = 0$:

$$\boxed{\ddot{\theta} + \tfrac{3R}{7m}\dot{\theta} + (\tfrac{27k}{7m} - \tfrac{12g}{7l})\theta \approx 0}$$

The natural frequency $\omega_n$ and damped frequency $\omega_d$ can be expressed (upon inspection) as follows:

$$\boxed{\omega_n = \sqrt{\frac{27k}{7m} - \frac{12g}{7l}}} \quad \text{and} \quad \boxed{\omega_d = \sqrt{\frac{3k}{7m} - \frac{12g}{7l} - \left(\frac{27R}{14m}\right)^2}} = \sqrt{\frac{27k}{7m} - \frac{12g}{7l}}\sqrt{1 - \left(\frac{3R/(14m)}{\sqrt{\frac{27k}{7m} - \frac{12g}{7l}}}\right)^2}$$

Just as before, this implies that the damping ratio can be expressed as follows:

$$\boxed{\zeta = \frac{3R}{14m\sqrt{\frac{27k}{7m} - \frac{12g}{7l}}}}$$

In the case that gravity is neglected, these equations reduce to:

$$\boxed{\omega_n = \sqrt{\frac{27k}{7m}}} \quad \text{and} \quad \boxed{\omega_d = \sqrt{\frac{27k}{7m} - \left(\frac{3R}{14m}\right)^2}} \quad \text{and} \quad \boxed{\zeta = \frac{R}{2\sqrt{21km}}}$$

Plenty could be said here, but if anything, neglecting gravity increases the chances the system oscillates (i.e. avoids over damping). Otherwise, gravity reduces $\omega_d$ when applicable.

Moreover, the systems in parts (a) and (b) are essentially identical relative to the structure of their EOMs (untampered/linearized). The nuance comes down to the weightings on the parameters $k, R, l, m$ with gravity ($g$) being slightly more influential in the latter when considered. For example, consider the weighting on $k/m$ with respect to $\omega_d$ in each case:

$$\left.\frac{\partial \omega_d}{\partial\left(\frac{k}{m}\right)}\right|_{(a)} = \frac{2}{9}\,\omega_d^{-1}\big|_{(a)} \quad \text{and} \quad \left.\frac{\partial \omega_d}{\partial\left(\frac{k}{m}\right)}\right|_{(b)} = \frac{27}{14}\,\omega_d^{-1}\big|_{(b)} \quad \implies \quad \frac{\left.\frac{\partial \omega_d^2}{\partial(\frac{k}{m})}\right|_{(a)}}{\left.\frac{\partial \omega_d^2}{\partial(\frac{k}{m})}\right|_{(b)}} = \frac{28}{243}$$

One can inspect that $\omega_d$ depends more on $k/m$ is case (b) relative to case ($a$). And so on so forth for the other weightings/parameters.